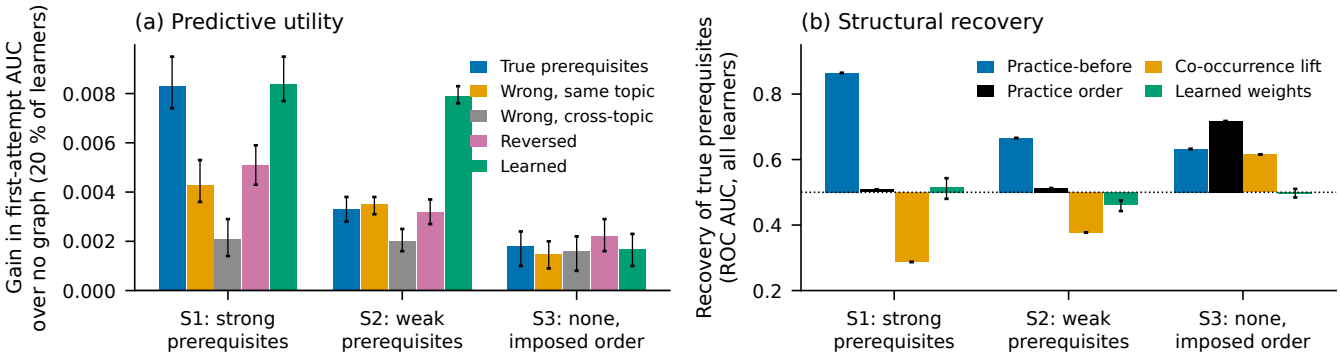
(a) Predictive utility
Gain in first-attempt AUC over no graph (20 % of learners)
0.000
0.002
0.004
0.006
0.008
True prerequisites
Wrong, same topic
Wrong, cross-topic
Reversed
Learned
S1: strong prerequisites
S2: weak prerequisites
S3: none, imposed order
(b) Structural recovery
Recovery of true prerequisites (ROC AUC, all learners)
0.2
0.4
0.6
0.8
Practice-before
Practice order
Co-occurrence lift
Learned weights
S1: strong prerequisites
S2: weak prerequisites
S3: none, imposed order